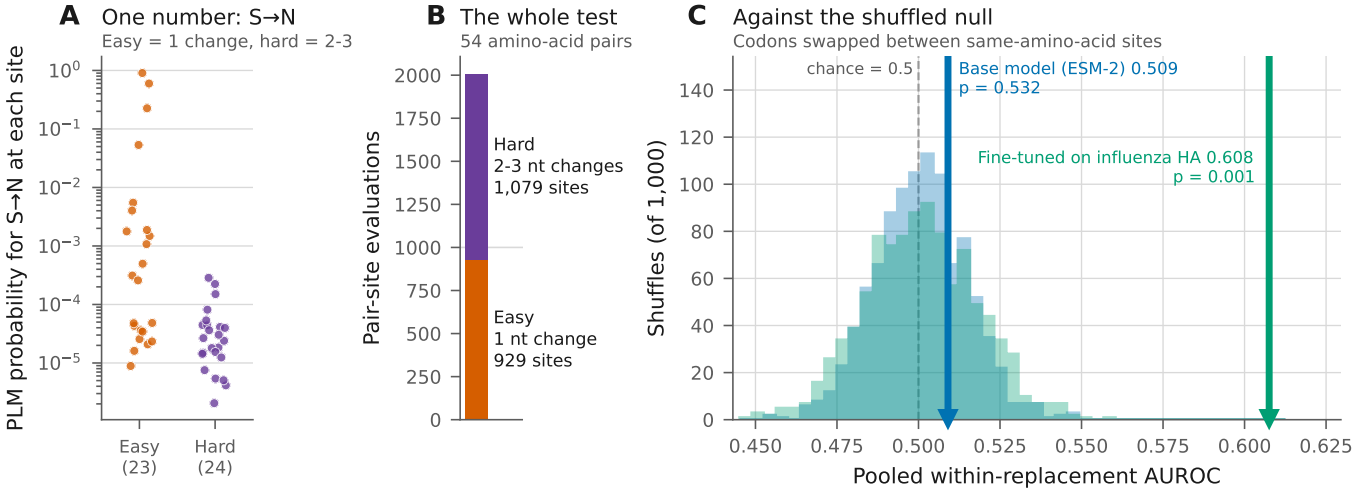
A
One number: S→N
Easy = 1 change, hard = 2-3
PLM probability for S→N at each site
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
Easy
(23)
Hard
(24)
B
The whole test
54 amino-acid pairs
Pair-site evaluations
2000
1750
1500
1250
1000
750
500
250
0
Hard
2-3 nt changes
1,079 sites
Easy
1 nt change
929 sites
C
Against the shuffled null
Codons swapped between same-amino-acid sites
chance = 0.5
Base model (ESM-2) 0.509
p = 0.532
Fine-tuned on influenza HA 0.608
p = 0.001
Shuffles (of 1,000)
140
120
100
80
60
40
20
0
0.450
0.475
0.500
0.525
0.550
0.575
0.600
0.625
Pooled within-replacement AUROC